INTRODUCTION TO ECONOMICS MGT404

Adverse Selection

Yale

Review and Today's Plan

Last Class

Making decisions based on expected value does not capture the desire to avoid risk
Risk-aversion can be modeled with expected utility, where the utility function has positive, but decreasing, marginal utility.
To have an idea of how much better one prospect is than another, we must turn to **certainty equivalent** and the **risk premium**.
A risk-averse person will only fully insure when premiums are actuarially fair. When there is a load, they will partially insure.

Today

Situations where parties do not have perfect information
Today: Adverse selection (hidden "type")

- One side of a transaction might have information that the other side does not have.
- There may be a distribution of "types" of people in the population
- If I offer a product or service, I may only attract one particular "type"
- The higher the price, the more likely to attract only those that are more costly to serve.

Next class: Moral hazard (hidden action)

Today’s plan: Adverse selection

Asymmetric Information and Adverse Selection

Suppose one side of a market has more information than the other. We should expect certain types to behave differently based on their information. This can lead markets to unravel.

Market equilibrium with adverse selection

- Firm’s average cost is a function of price, since price determines the types of customers who buy.
- Equilibrium is possible, where supply and demand meet despite adverse selection. However, not guaranteed.

Market Failure: Death Spirals

- Healthcare: if mostly sick people purchase insurance, premiums rise, so even more healthy people drop out of the market.

Solutions for adverse selection

- Buying information
- Widening the surplus gap
- Signaling
- Screening
- Commitment devices (reputation, legal system)

The Market for Lemons

The market for the N 2021 Honda Civics potentially for sale in Connecticut:

80% of the Civics that might potentially be up for sale are **"peaches"**

- Potential buyers willing to pay $11,500 if they know they're getting a "peach."
- Sellers have willingness to sell somewhere between $8,000 and $12,000

20% of the Civics are **"lemons":** (they will need a $2,500 repair soon)

- Buyers value at $9,000; all sellers willing to sell at $8,000

HIDDEN INFORMATION: Sellers know what sort of Civic they have, but buyers are incapable of distinguishing lemons from peaches at the time of purchase.

If all used Civics are selling, then a for-sale Civic has an expected value to buyers of $20\% \cdot \$9\text{K} + 80\% \cdot \$11.5\text{K} = \$11\text{K}$.

At this price, some Civic owners won't sell…

…which pushes the expected value of a for-sale Civic down…

…which makes even fewer Civics owners want to sell…

and so on…

Does the market completely *unravel*?

How efficient is this market?

Demand for Used Cars

First, let's look at demand for used cars:

Assumptions:

Buyers are Risk Neutral

Buyers Know the mix of Lemons/Peaches for Sale

Buyers would only buy a used car if the price is less than or equal to their expected value for a used car.

We can define the **demand** for used cars as the maximum price customers are willing to buy for a Honda Civic as a function of ***f* – *the fraction of peaches that sell.***

Demand for used cars

Suppose **all** Lemon owners are selling their cars for anything above $8,000, and some fraction f of peach owners are selling theirs.

Expected Value of Buying a Used Car

Suppose **all** Lemon owners are selling their cars for anything above $8,000, and some fraction f of peach owners are selling theirs.

$$p(f) = EV = \frac{20\% \cdot 9{,}000 + 80\% \cdot f \cdot 11{,}500}{20\% + 80\% \cdot f} = \frac{1{,}800 + 9{,}200f}{0.2 + 0.8 \cdot f}$$

This gives us a maximum price for each f at which buyers are willing to buy a used car.

Supply of Used Cars

Reservation Price for Peach Owners to Sell

Suppose that owners of "peaches" differ in their willingness to sell: reservation prices at which an owner would sell are distributed uniformly between $8,000 and $12,000.

For a market price of p between $8,000 and $12,000, the fraction selling is:

$$f(p) = \frac{p - 8000}{4000}$$

Sales increasing in price from 8k

12000-8000

$$p(f) = 4000f + 8000$$

Market Equilibrium? Look for a Price where Supply Equals Demand:

$$4000f + 8000 = \frac{1{,}800 + 9{,}200f}{0.2 + 0.8 \cdot f}$$

$$3200f^2 - 2000f - 200 = 0$$

$$16f^2 - 10f - 1 = 0$$

Supply of Used Cars

Market Equilibrium? Look for a Price where Supply Equals Demand:

$$16f^2 - 10f - 1 = 0$$

Reminder: Solution to $ax^2 + bx + c = 0$ is

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

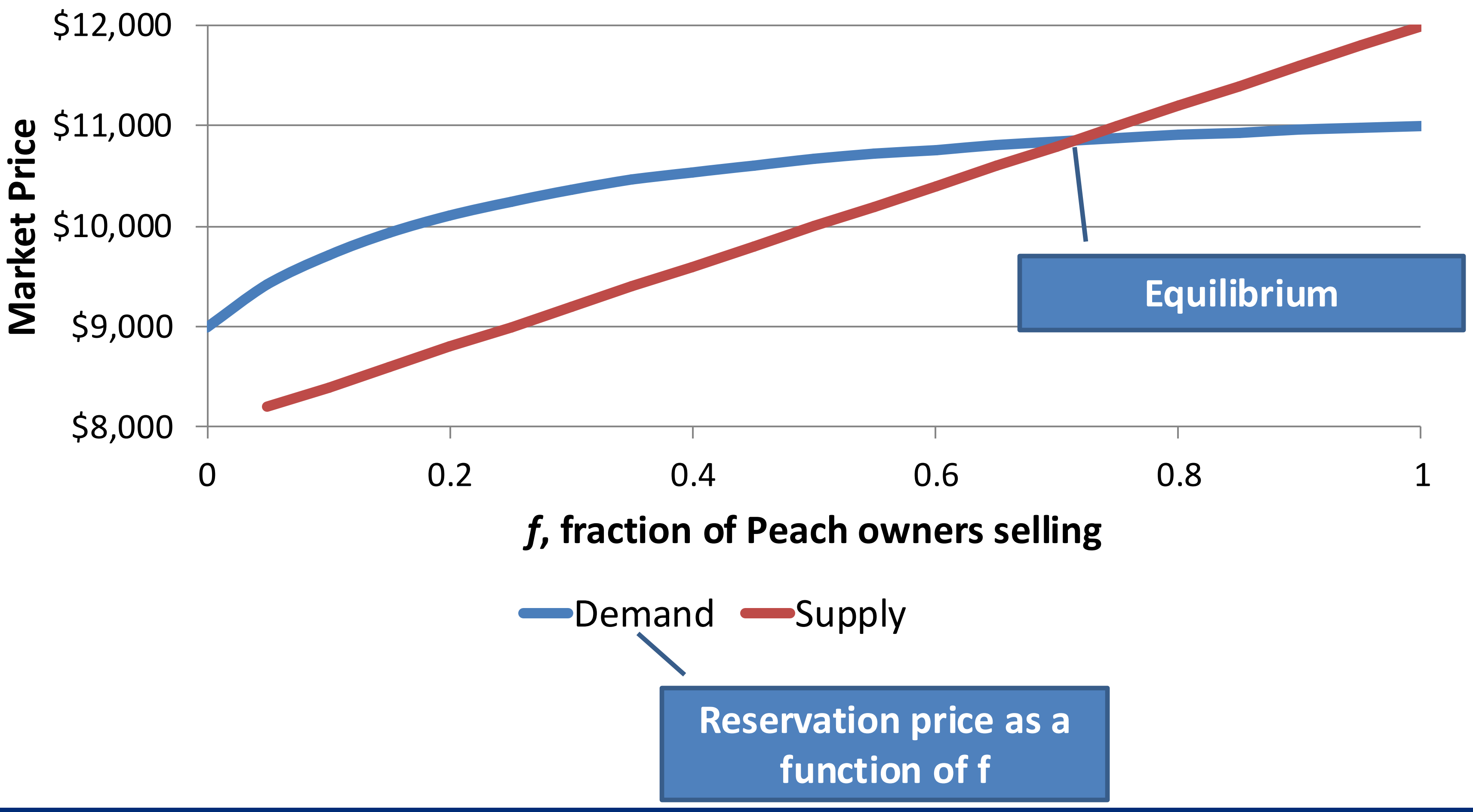
The positive solution to our equation is f = 0.713. The corresponding price is $10,851.

At a market price of $10,851, about 71.3% of peach owners are willing to sell, and buyers are willing to pay that price for a chance of a lemon.

In equilibrium, 0.2/(0.2+0.713 *0.8) = 26% of sold cars will be lemons.

- What if the market price were higher? Lower?

Graph of Outcome



Discussion of Market Outcome

How good of a market outcome is this? What should we compare it to?

Suppose there were Perfect Information ("First Best")

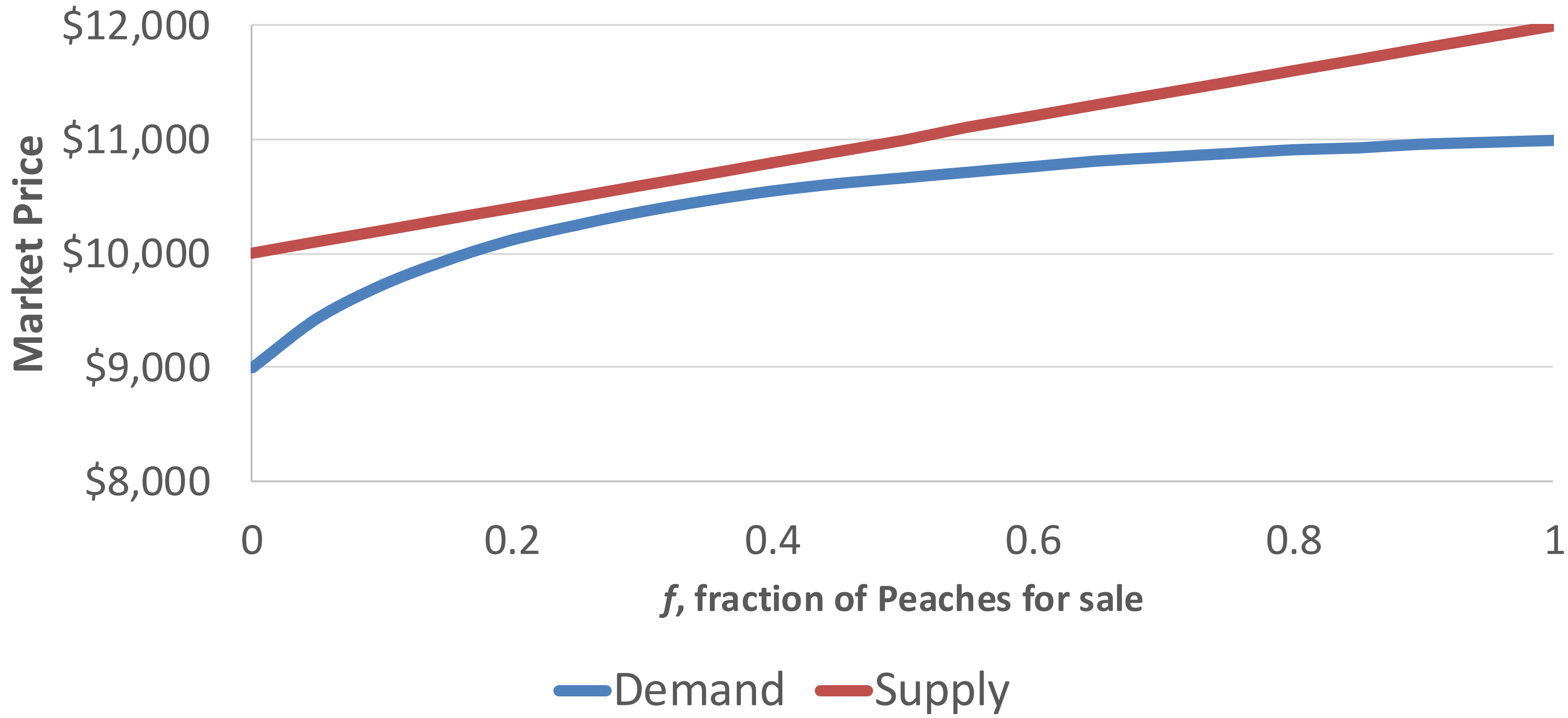
- In this setting, there are always gains from trading "lemons" (sellers' values=8,000; buyers' values-9000)
- In contrast, all "peach" sellers with values below 11,500 (87.5%) should sell.

In our equilibrium, only 71.3% of "peach" sellers are willing to sell.

- There are gains from trade that are not being realized.
- Asymmetric information has "cost" the market some surplus.

Graph of Outcome: No Equilibrium

Now: peach owners have reservation prices to sell uniform on $10,000-$12,000:



There is no equilibrium price anymore. Any car for sale will be assumed to be a lemon. Today's class will cover some strategies to improve how markets function when there is asymmetric information.

Today’s plan: Adverse selection

Asymmetric Information and Adverse Selection

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Market Failure: Death Spirals

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Adverse Selection and Market Failure

The Problem: Adverse Selection

Health Care Insurers are allowed to offer different prices under certain circumstances; the policy for a 60 year-old is not going to have the same price as for a 20 year-old. However, within demographic groups, prices are effectively the same.

How does this affect the market for health insurance?

Set an insurance premium, $P.

Only those with an expected cost above $P would purchase.

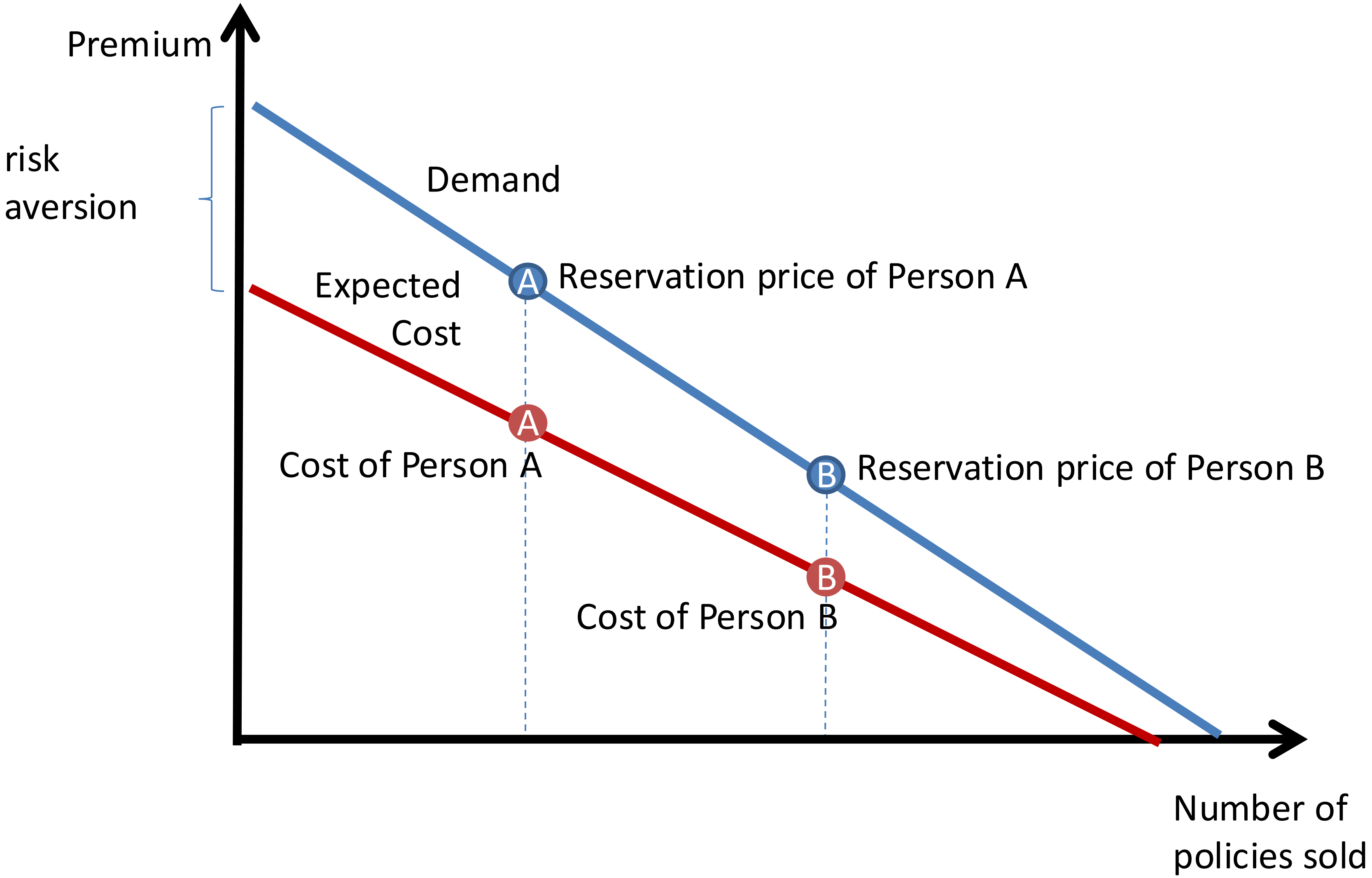
Therefore, average cost will be above $P.

What does this mean for insurance companies?

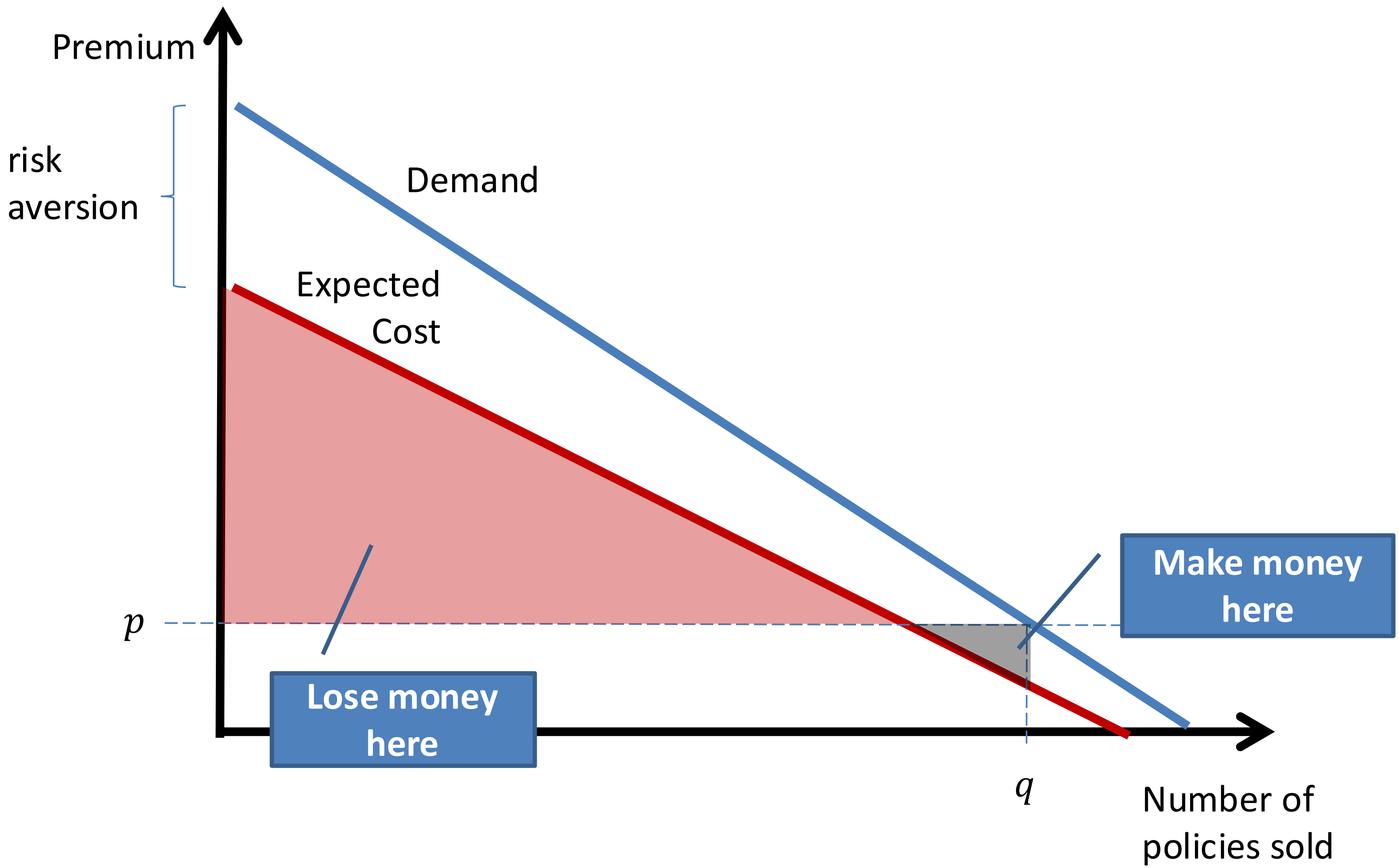
This can result in a *death spiral*. The ACA has hopes to address this with a *mandate*.

Plotting Demand and Supply in Health Care

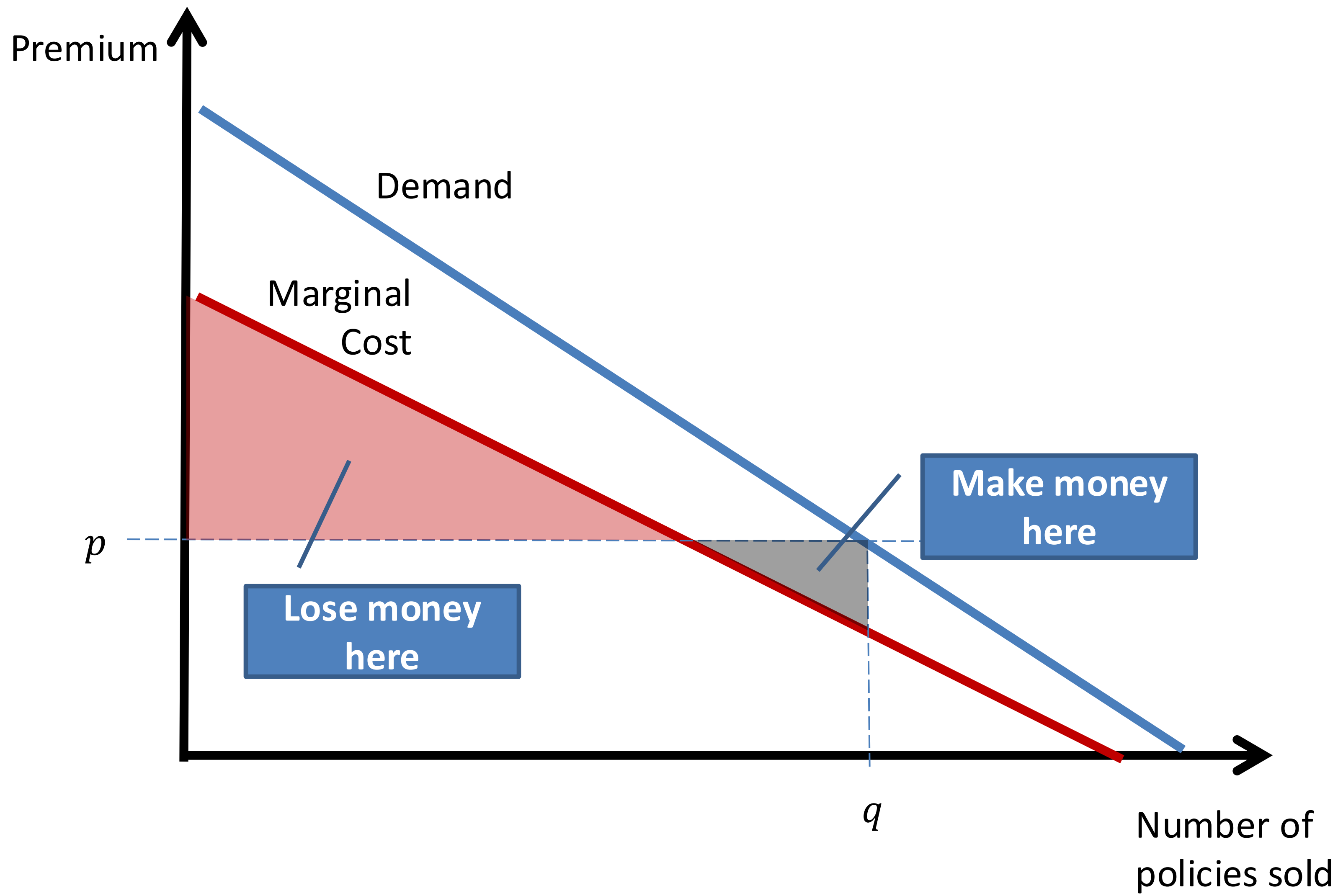
Unlike before, different consumers **cost** more or less to the firm, depending on how much health care they will use.



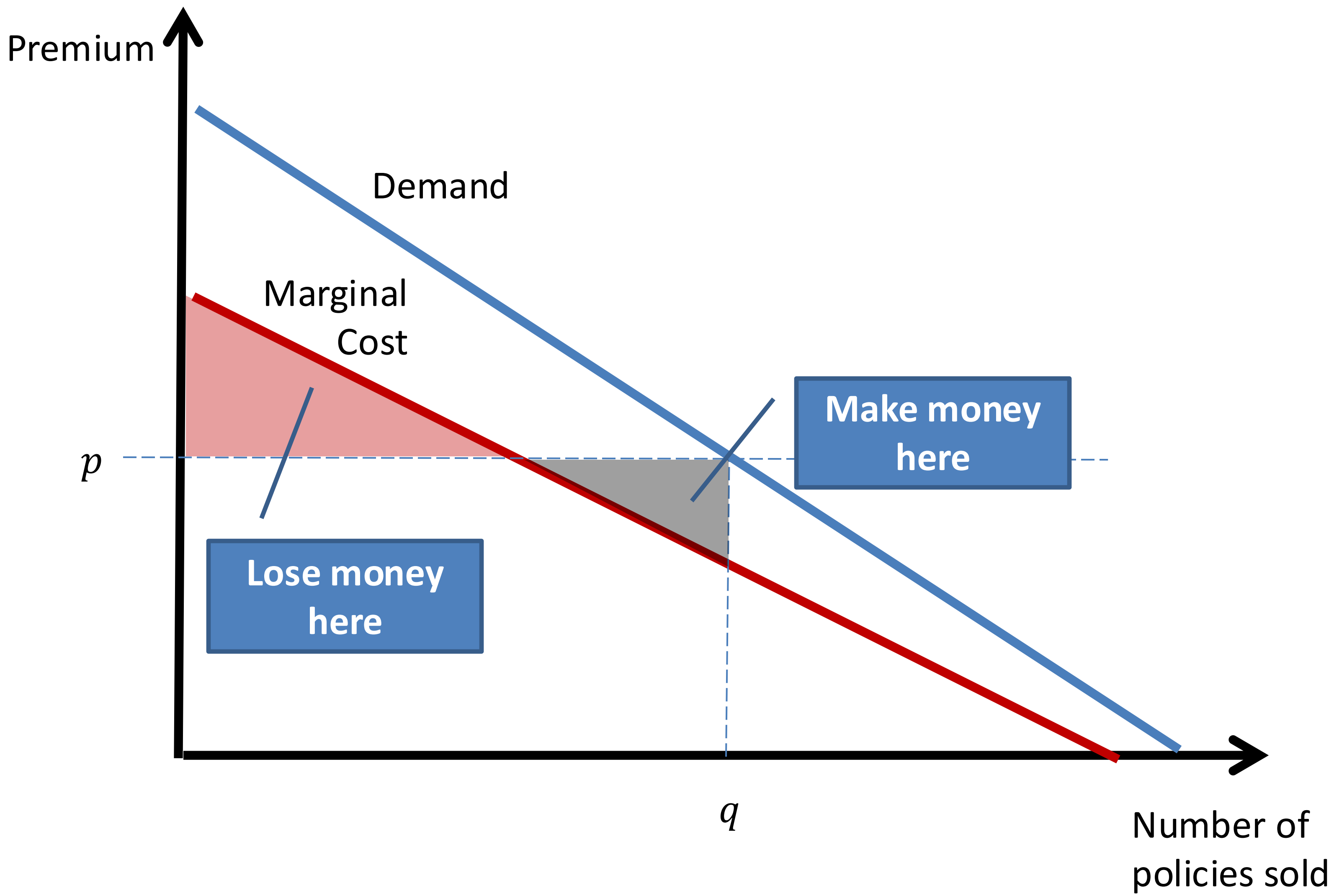
Equilibrium Premium?



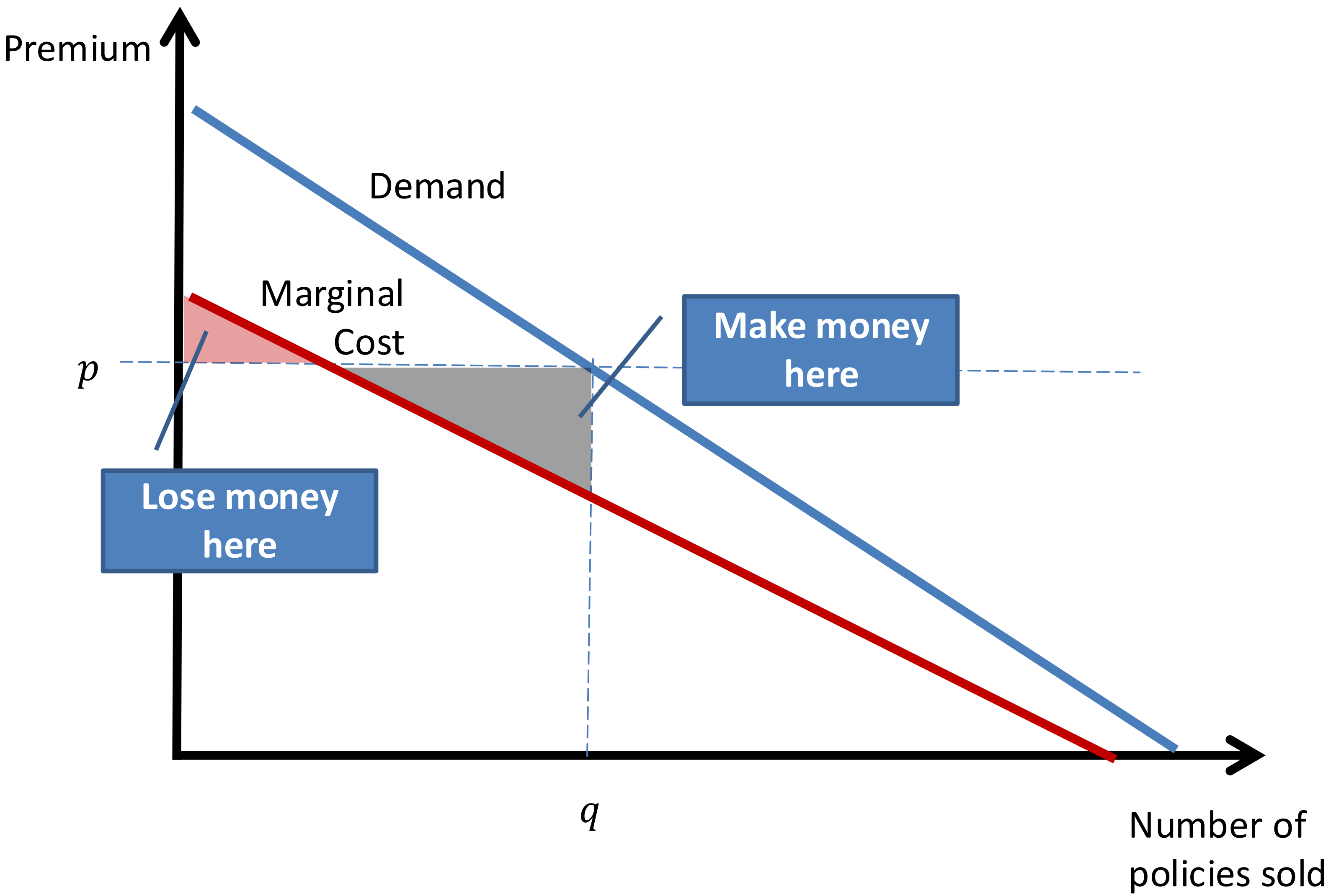
Equilibrium Premium?



Equilibrium Premium?



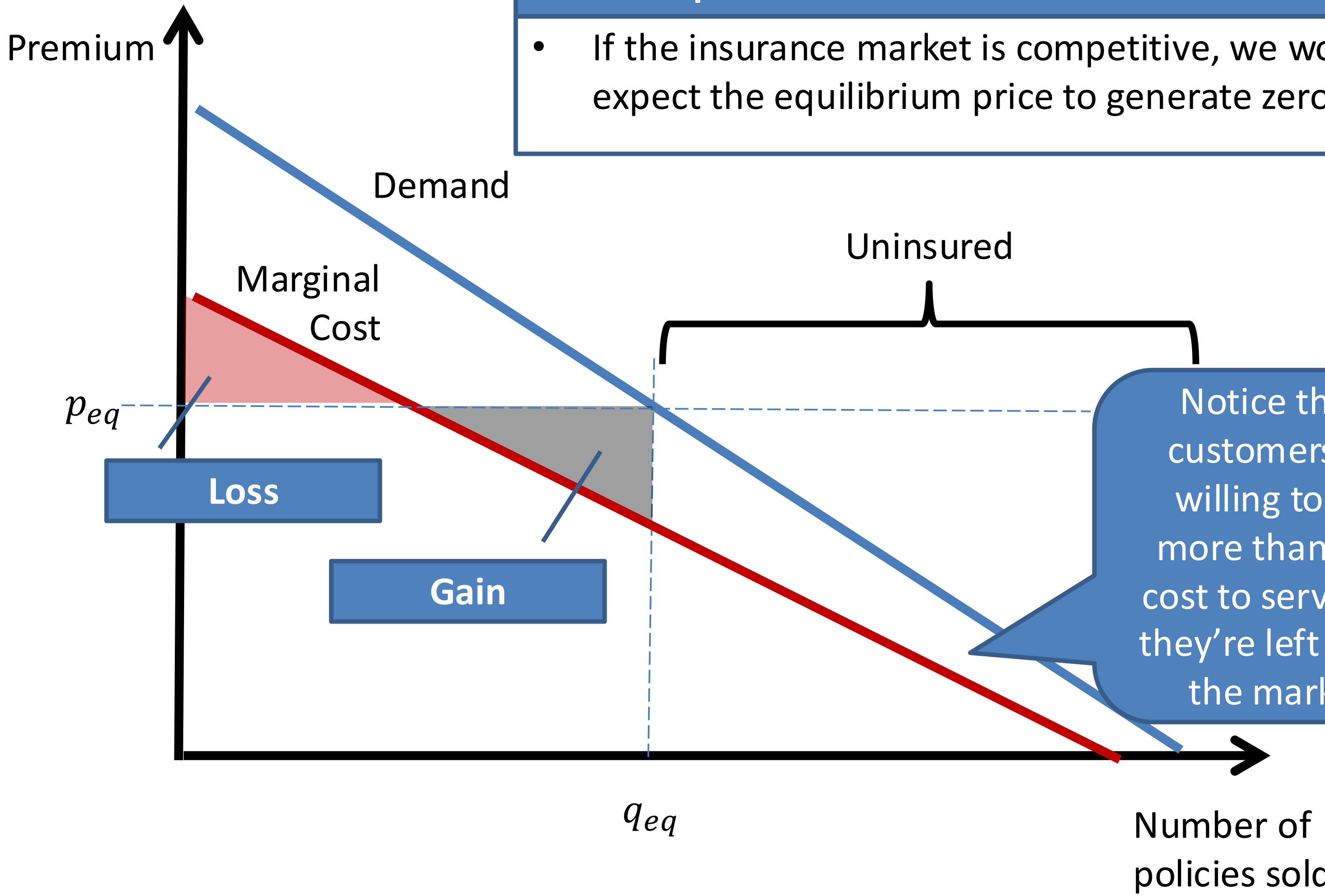
Equilibrium Premium?



Equilibrium Premium?

Market Equilibrium

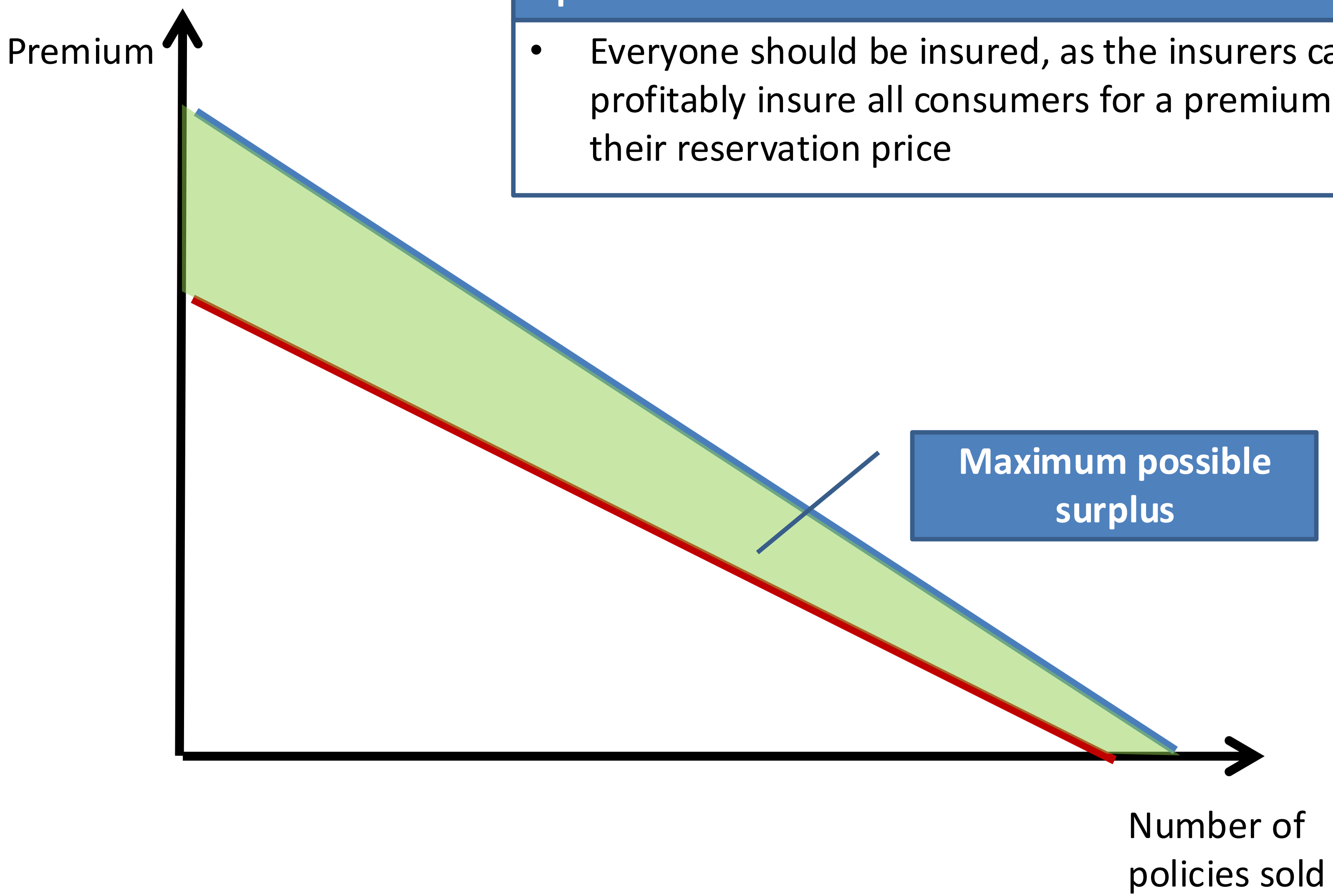
- If the insurance market is competitive, we would expect the equilibrium price to generate zero profits.



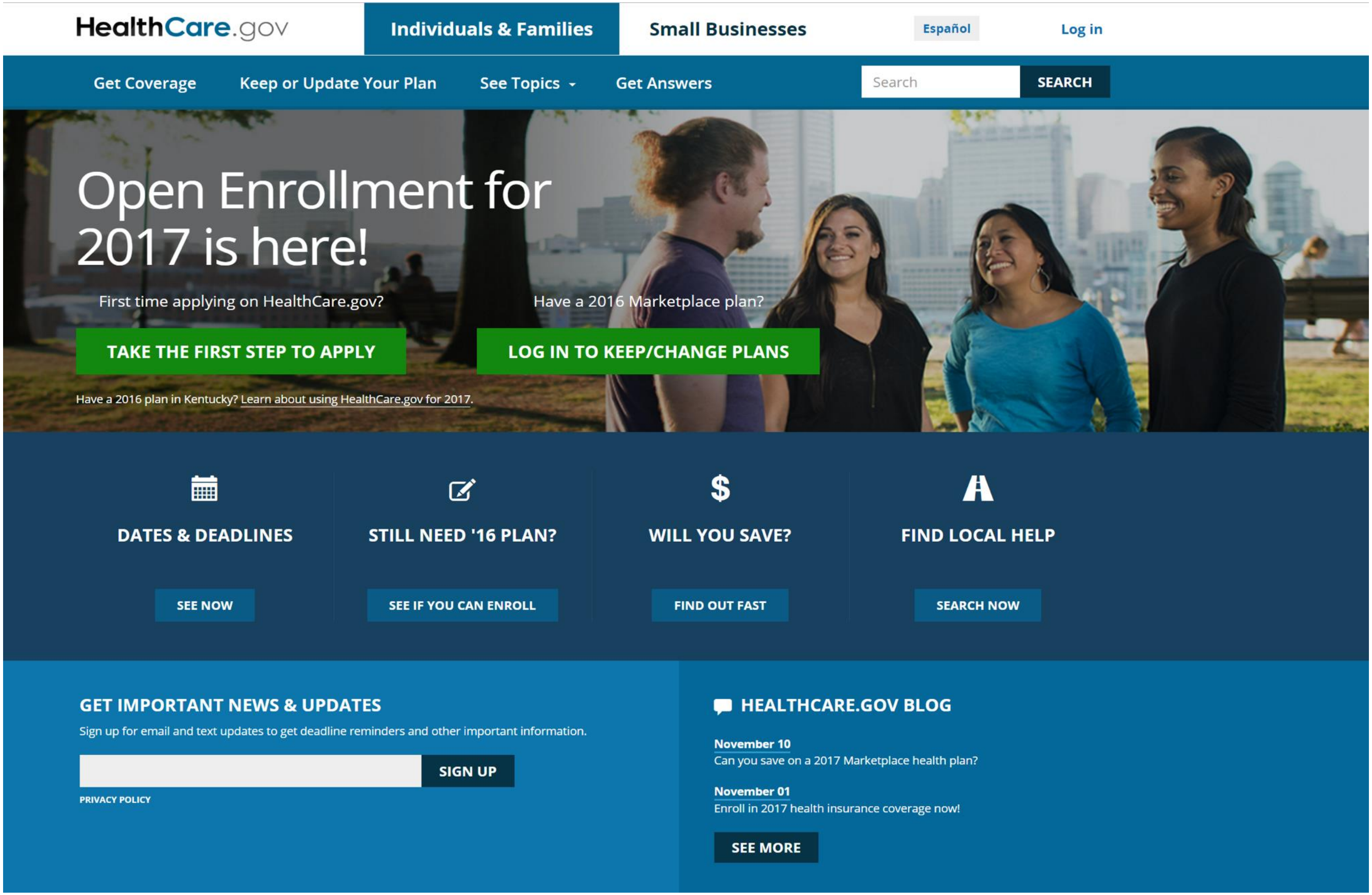
Market Failure

Optimal Outcome

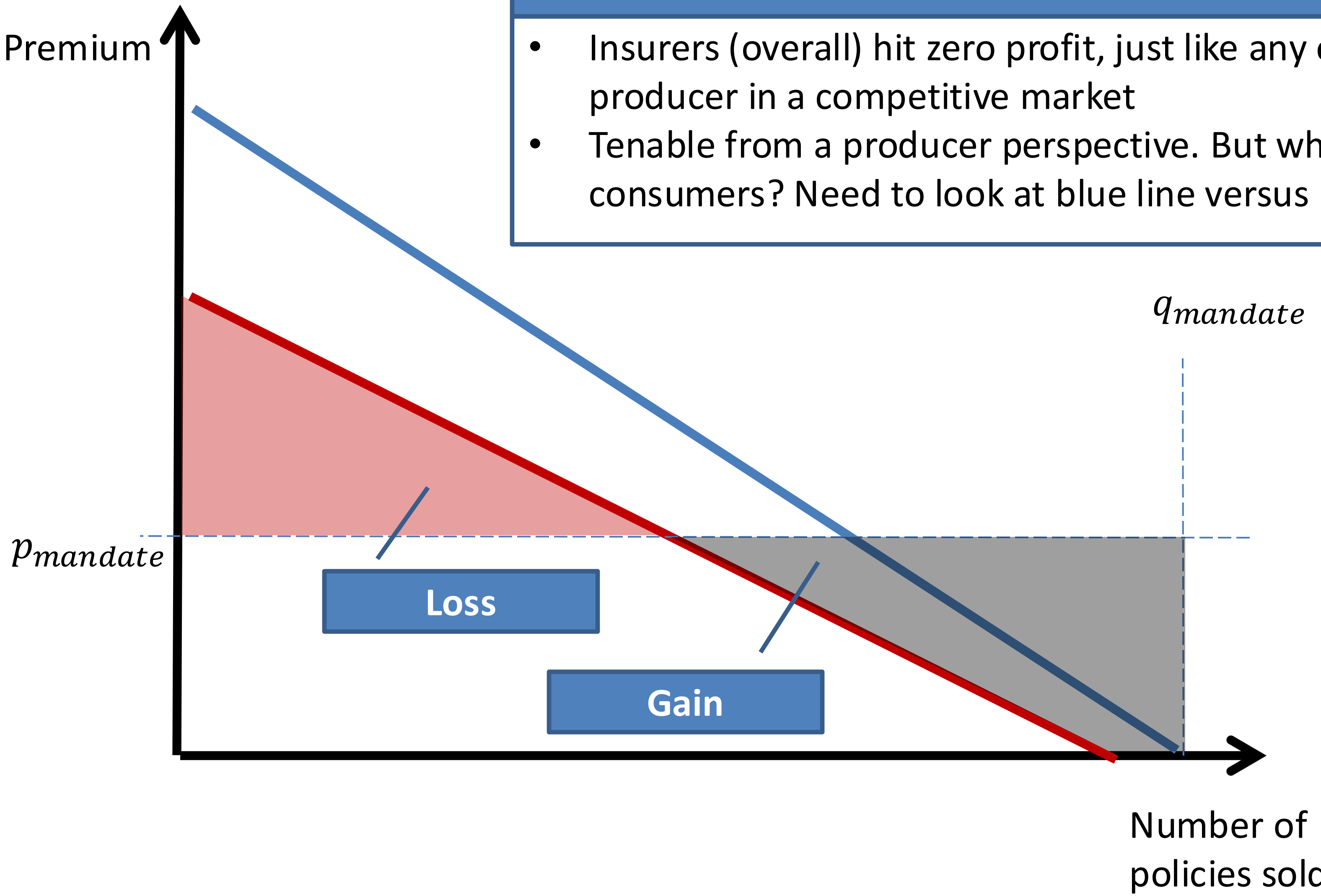
- Everyone should be insured, as the insurers can profitably insure all consumers for a premium below their reservation price



Affordable Care Act



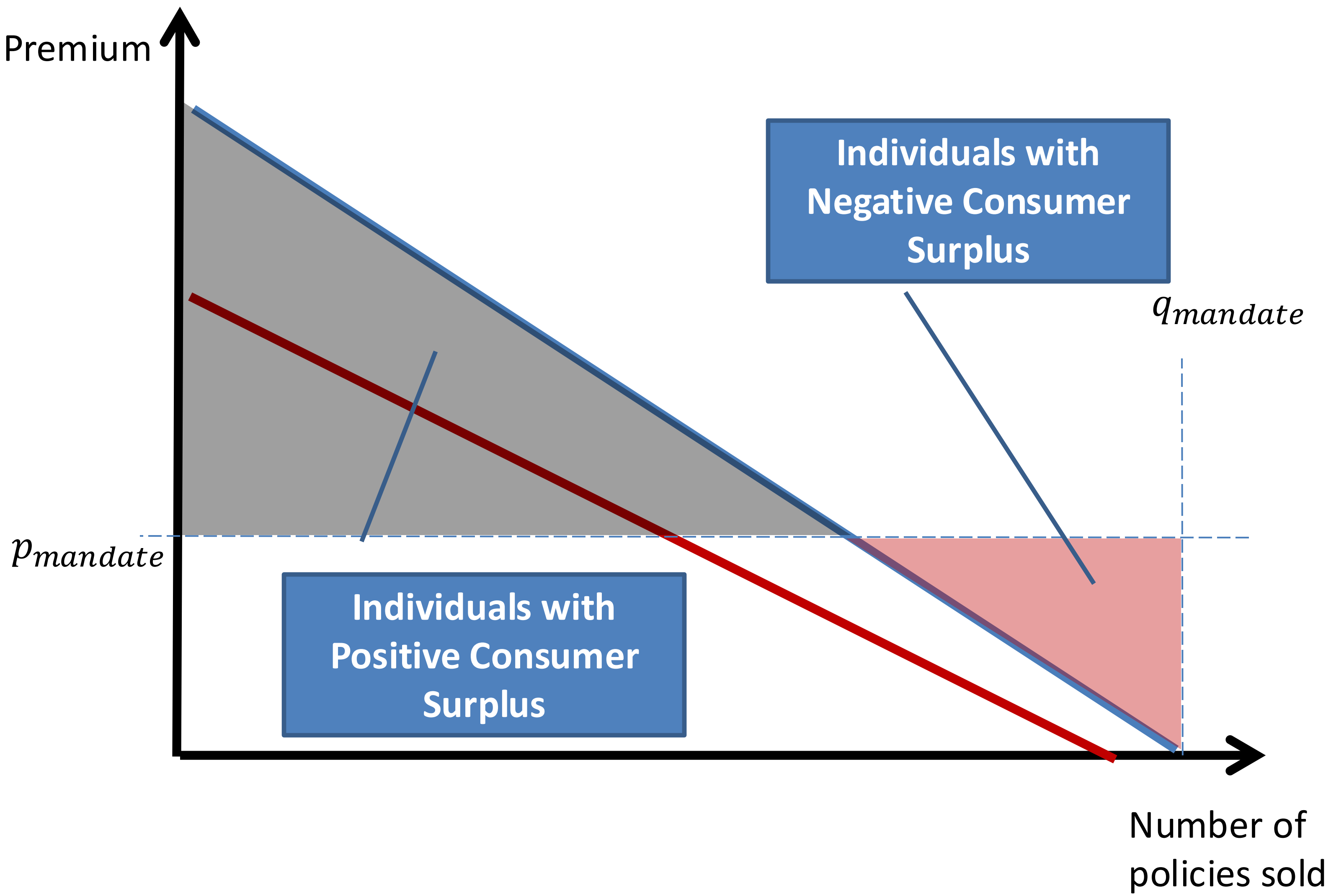
Healthcare: Mandatory Insurance?



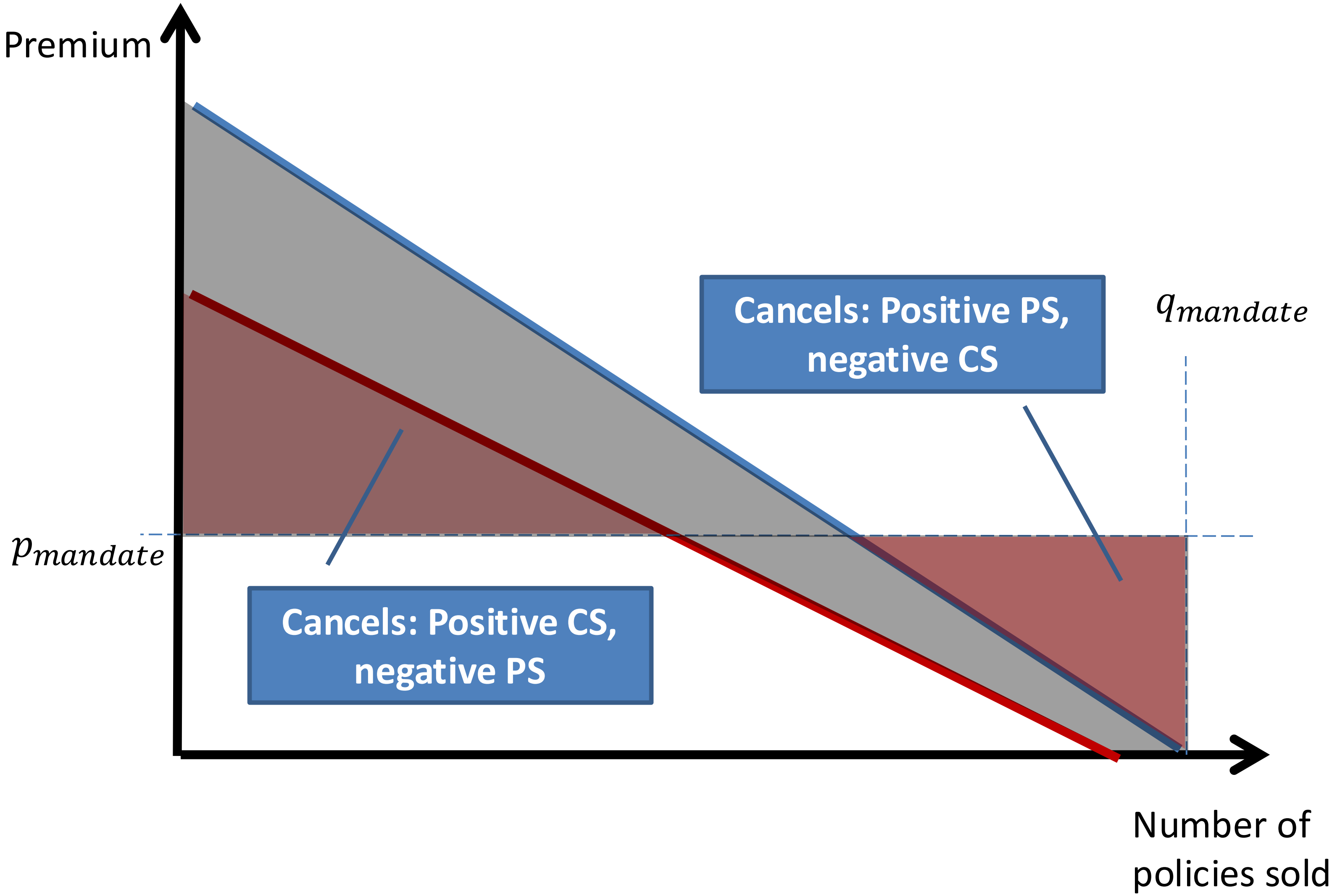
Introduce a Mandate

- Insurers (overall) hit zero profit, just like any other producer in a competitive market
- Tenable from a producer perspective. But what about consumers? Need to look at blue line versus price.

Why Such Opposition? *Consumer Surplus*



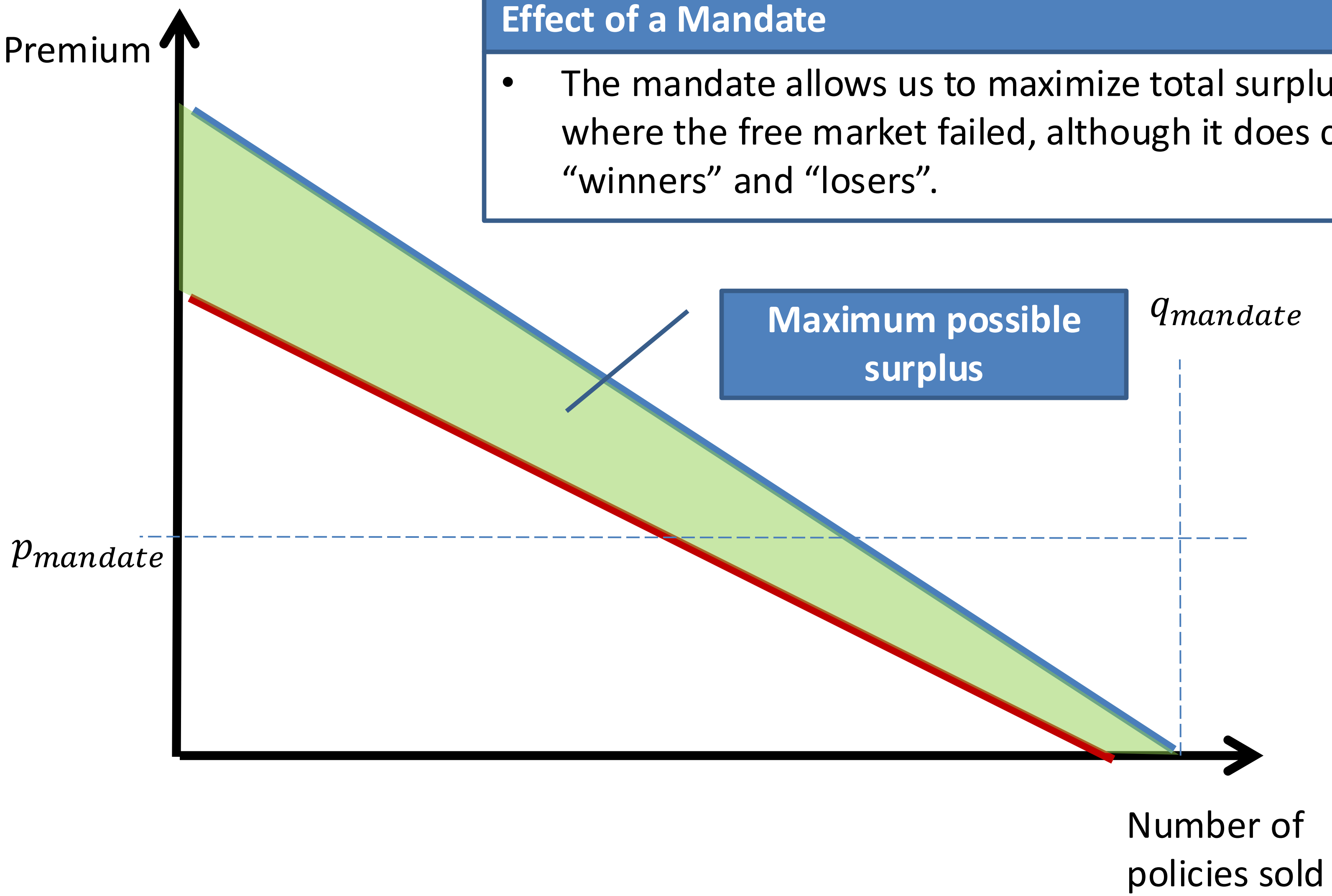
Total Surplus?



Market Failure Solved

Effect of a Mandate

- The mandate allows us to maximize total surplus where the free market failed, although it does create "winners" and "losers".



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Overcoming adverse selection

A playbook of strategies for profiting in markets with adverse selection

1	**Buying information**	I buy information about my transaction partner from a third party. (e.g., get a mechanic to check out the car)
2	**Increase the surplus gap**	If transactions create more surplus, then the unraveling caused by adverse selection may be mitigated if low cost customers can subsidize high cost.
3	**Signaling**	I credibly signal my “type” with something only someone of my type could do (It’s credible because the “bad” type can’t “afford” to send the same signal).
4	**Screening**	I set up a menu that makes types self-select into pricing schemes. This is similar to creating self-selecting price discrimination schemes.
5	**After-the-fact recourse/reputation**	If I am punished for taking advantage of my informational advantage, I might not do so. Usually, this works through reputation, but sometimes it works through the legal system (fraud and disclosure laws).

Just for references from now on…
Not to be tested in the exam

1 "Buying" Information: Due Diligence

Suppose Mechanics could Tell Lemons from Peaches

Suppose that there were mechanics in the market that were able to inspect a car and correctly determine whether or not a car was a "peach" or a "lemon". How might this affect the market?

Mechanic Cost:	Outcome:
Mechanics are Free	Buyers would observe the truth; no more asymmetric information. • Peaches will sell for \$11,500, 87.5% of owners would sell • This is the *first-best* outcome.
Mechanics charge \$C	Buyers could hire mechanics: • Peaches will sell for \$11,500-\$C, a fraction $\frac{3500-C}{4000}$ of owners would sell. • This requires \$C < (\$11,500-\$10,851)=\$649 to have any transactions. *Market clearing p*

Buying information can correct part of the problem.
This creates an industry of "valuers"

1 “Buying” Information: Due Diligence

Discussion questions

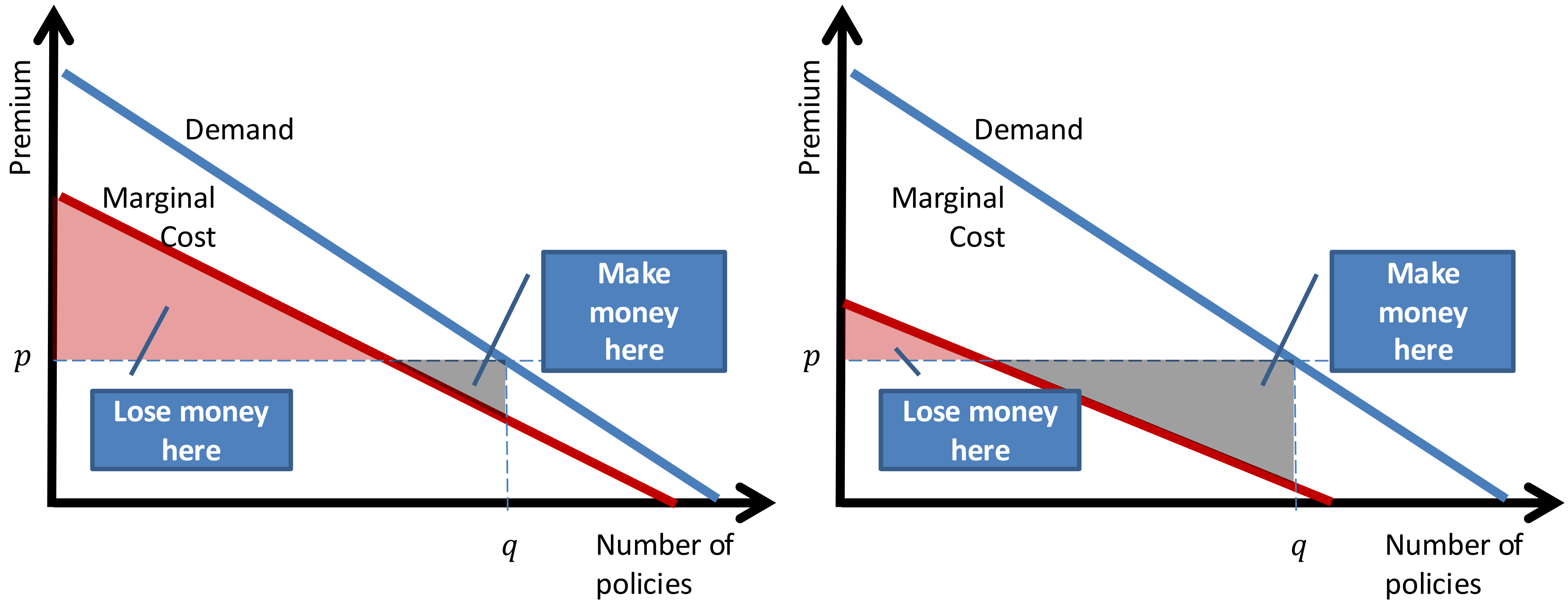
- Why does the buyer “pay for” the mechanic?
- Does the buyer really pay the entire price of the mechanic?
- Qualitatively, how many transactions will there be if mechanics charge a positive price?
- Are the mechanics necessarily honest? What are the incentives here?
- What features of markets generate valuers?

Examples of “valuers”

- Carfax
- Appraisers for homes, art, jewelry.
- Antique Roadshow
- Due diligence, consultants.
- Bond rating agencies (Moody’s, S&P, etc)
- Others?

2 Increasing the surplus gap

- What kind of market won't unravel at all?
  - You can imagine a case where the buyer is willing to pay so much above the seller's reservation value that all peach owners are willing to sell, keeping average quality high
  - Or in a firm/customer model, "bad" types cost more, but not enough that they erase profits—the "good" types are willing to pay enough that they can subsidize the "bad types.



2 Increasing the surplus gap

Can I decrease costs (especially for high types), or increase value to high types without increasing costs?

Examples of "changing the surplus gap"
- Buffets (design product so that costs are low) - Slowing down data speed at high levels of consumption - Others?

Do all-you-can-eat buffets suffer from adverse selection?
Why do they continue to exist?
Who eats at an all-you-can-eat buffet? What are costs like?
How might I design my restaurant differently?

Signaling with Warranties

Suppose Sellers could Offer Warranties

Suppose a seller could set a price $p_w > p$ that includes a warranty that covers w dollars of repairs. Could this correct the market?

Seller Profits

	Peach Owner	Lemon Owner
Seller Profits	• p_w if warranty included • p if no warranty	• $p_w - w$ if warranty included • p if no warranty.
	Knows it's a peach, happy to provide any warranty	Will provide a warranty if $p_w - w > p$

Warranty Equilibrium

Peach owners could offer a $2,500 warranty and set a price of $11,500. Peaches would sell, but lemons would only be sold without a warranty at $9,000. **Lemon owner not willing to provide warranty. It is a credible signal.**

Signaling Examples

Examples of Signaling:

Money-Back Guarantee

If you can learn the true quality after use, this makes sense. However, could be abused.

- Perhaps include a restocking fee?

Product Reviews

Magazines review products all the time. Problems?

- They sell ads to firms; maybe they lie?
- Consumer Reports: no ads. Sued for Libel on many occasions, always cleared and awarded attorney fees.

Credentials

How important are professional designations to getting a job? Some carry more weight than others. If a person has an MBA, does that signal work ethic and intelligence?

Dividends

If dividends are tax-disadvantaged, why offer them?

- If I need to retain earnings to finance my investments, am I weak?

This phenomenon is pervasive.

4 Screening customers: auto insurance

Sellers Cannot Always Distinguish between Buyers of Different Types

The canonical example of adverse selection is insurance: a buyer knows more about their expected cost to an insurer than the insurance company. How does this play out?

Suppose there are two types of drivers in New York:

Safe & Courteous

Probability of an accident this year is 1%

- Average wealth of $250K, risk-averse with $\ln(x)$ utility.

Reckless

Probability of an accident this year is 5%

- Also: average wealth of $250K, risk-averse with $\ln(x)$ utility.

Suppose there are equal types of each in the market and that **an accident would cost $100K**.

Therefore, expected cost of insuring a Safe driver is $1,000, and a Reckless driver is $5,000.
Recall, these individuals will decide based on expected utility.
How will the car insurance market work?

4 Auto Insurance screening

Suppose AllState **cannot** tell which type of driver you are. One idea is to price to the average cost in the market (here, this is \$3,000).
Who will purchase insurance now?

Purchase Decision:

Reckless

Expected utility insured: $EU = \ln(250K - 5K) = \boxed{12.409}$

Expected utility uninsured: $EU = 0.05\ln(150K) + 0.95\ln(250K) = 12.404$

Safe & Courteous

Expected utility insured: $EU = \ln(247K) = 12.417$

Expected utility uninsured: $EU = 0.01\ln(150K) + 0.99\ln(250K) = \boxed{12.424}$

So, only "Reckless" will purchase…

- … so the average cost goes up…
- … so the price goes up…
- … until only "Reckless" are buying insurance, and the price is back to \$5,000.

How could the insurers fix this?

Auto Insurance screening

4

Suppose AllState Offered Two Insurance Plans…

Consider the option of offering two different insurance packages:

- Plan 1: Premium of $5,000, no deductible.
- Plan 2: Premium of $100, deductible of $90,000.

Which plan would each type purchase? (Note, this is an extreme example)

	EU of "Safe"	EU of "Reckless"
Plan 1 (Full Ins.)	$\ln(245K) = 12.409$	$\boxed{\ln(245K) = 12.409}$
Plan 2 (Partial Ins.)	$\boxed{.01\ln(159.9K) + .99\ln(249.9K) = 12.42435}$	$.05\ln(159.9K) + .95\ln(249.9K) = 12.406$
No Insurance	$.01\ln(150K) + .99\ln(250K) = 12.42411$	$.05\ln(150K) + .95\ln(250K) = 12.404$

"Safe" buys Plan 2, "Reckless" buys Plan 1. Both insure, at least partially.

These self-selecting plans screen customers, reducing adverse selection problem

4 Outcome with Screening

Screening As a Strategy

A firm should always do its best to segment (“screen”) customers based on cost.

- What if you don’t? What if you serve a broad base at average cost?

You may leave open an opportunity for competitors to “cream skim” your best customers.

If my low-cost customers are subsidizing my high-cost customers,

- A competitor may be able to steal my low-cost customers
- …which would destroy my business model!

Even if a market doesn’t unravel, you risk leaving open an opportunity for competitors to “cream skim”—so serving each customer with unique plans, via self-selecting screening, is better.

5 Reputation and legal recourse

- One last way to overcome adverse selection: make promises about "type" enforceable. Either:
  – Have reputations, so people won't lie about their type, because of repeated game (e.g., trusted auto dealers that will set price according to value)
  – Have a legal system that will enforce promises about type (e.g., you can sue someone after the fact if you find out they sold you a "lemon")

- With these "after the fact" corrections in place, there is no benefit to lying, and so agents will reveal their types, and be priced accordingly. (This eliminates the efficiency loss from adverse selection)

Examples of reputation and legal recourse

- Airbnb
- Uber
- Branding as reputation
- Others?

Summary and Next Lecture

Today

Adverse selection is when one party is able to conceal their type, which increases costs (or, in the car case, lowers value) creating a pool that is *adversely selected.*

Strategies for overcoming:

- **INCREASE THE SURPLUS GAP**: The bigger the difference between cost and willingness to pay, the more "good" types can subsidize "bad"
- **BUYING INFORMATION:** Even when very costly, information can be worth it to acquire
- **SIGNALING**: Seller/customer can signal in a way that low-quality types will not replicate
- **SCREENING**: Firm sets up a menu of options that people self-select into
- **REPUTATION**: Repeated games (or legal system) can sustain higher equilibria

Next Class

Moral hazard (hidden action)

- Sometimes we cannot monitor/observe an action (or it is very costly)..
- How can we set incentives so that people behave in a way that we want them to?